Agentic AI for Robust and Socially Aware Autonomous Driving

Federated Collective Learning Between Autonomous Vehicles using Agentic Knowledge Sharing

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Project Proposal Report

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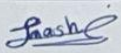
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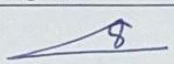
DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Autonomous vehicles mainly learn from their own driving experiences and usually operate independently from other vehicles. When one vehicle encounters a rare or unfamiliar situation, such as a newly introduced traffic sign or a temporary road condition, it learns from that experience locally. However, other vehicles in the fleet cannot immediately benefit from that knowledge. As a result, each vehicle may need to face and learn the same situation repeatedly, which slows down learning and may affect road safety. Although federated learning allows vehicles to collaboratively improve their models, most existing approaches share all updates without considering their importance, leading to unnecessary communication and sometimes unreliable information.

This research proposes an Agentic Federated Knowledge Sharing framework for traffic sign recognition using the German Traffic Sign Detection Benchmark (GTSDB) dataset. In the proposed approach, three vehicles initially trained on different traffic sign categories prohibitory, danger, and mandatory using YOLOv8. Each vehicle detects traffic signs and classifies observations such as KNOWN, RARE, or NEW based on confidence and frequency thresholds. An important score then calculated using detection confidence, bounding box size, and novelty level. A Deep Q-Network (DQN) reinforcement learning agent decides whether the detected knowledge should be shared with other vehicles.

Shared observations sent to a global verification model that validates the information and distributes verified knowledge to the fleet. This allows vehicles to learn from each other's experiences and improve their recognition of rare traffic signs.

The expected outcome is a prototype system that demonstrates selective knowledge sharing, reduced communication overhead, and faster fleet-wide learning, helping autonomous vehicles adapt more quickly to new traffic scenarios.

Keywords: Federated Learning, Reinforcement Learning, Autonomous Vehicles, Traffic Sign Recognition, Agentic AI, Deep Q-Network, YOLOv8, Novelty Detection

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This proposal reflects my individual contribution to the broader research project while benefiting from the guidance and collaborative efforts of my team.

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LIST OF ABBREVIATIONS

Abbreviation	**Definition**
AV	Autonomous Vehicle
RL	Reinforcement Learning
DQN	Deep Q-Network
GTSDB	German Traffic Sign Detection Benchmark
YOLOv8	You Only Look Once version 8 (Object Detection Model)
KB	Knowledge Base
KPI	Key Performance Indicator
CNN	Convolutional Neural Network
GPS	Global Positioning System
NFR	Non-Functional Requirement
FR	Functional Requirement

1. INTRODUCTION

1.1 Background and Context

Modern autonomous vehicles (AVs) operate as isolated learners, training on local data without leveraging fleet-wide intelligence. When encountering rare or unseen scenarios such as new traffic signs, temporary construction zones, or unusual weather AVs cannot benefit from experiences of other vehicles, leading to repeated learning failures, inefficiency, and compromised road safety [1].

This issue affects multiple stakeholders. AV drivers face safety risks from unrecognized traffic signs [1]; vehicle manufacturers incur high costs updating fleet models individually [2]; fleet operators experience operational disruptions when vehicles encounter unknown scenarios [3]; transportation authority's struggle to introduce new signage effectively [4]; and the public is impacted, as AV perception failures contribute to approximately 1.35 million annual road deaths globally [5]. A single AV generates around 4TB of data daily, yet traditional federated learning wastes 60-70% of communication bandwidth and requires over 2000 communication rounds to achieve 85% accuracy [6][1]. Globally, 247 new traffic sign types have appeared in the last decade, and in developing countries, 15-20% of signs are non-standard or temporary [7][8].

Existing autonomous vehicle learning systems have critical limitations. Traditional federated learning approaches share all updates indiscriminately [1][4][6], agentic knowledge sharing and communication protocols often cannot prioritize critical knowledge [2][3][8], and LLM-based or centralized solutions are computationally heavy or require full retraining for new signs [5][9][10]. Crowdsourced mapping, multi-vehicle coordination, edge computing, and continuous learning systems face slow updates, limited novelty detection, and catastrophic forgetting [1][2][3][4][5][6][11]. Collectively, these approaches fail to intelligently select what knowledge to share and cannot adapt in real time.

This proposed system addresses these gaps using an agentic federated learning framework for traffic sign recognition. A Deep Q-Network (DQN) agent evaluates the importance of detected observations based on confidence, novelty, context, and safety to decide whether knowledge should share. Traffic signs detected using YOLOv8, generating feature embeddings and metadata that form compact knowledge packets sent to a global verification model. The global model independently verifies observations using its own YOLOv8 predictions against GTSDB ground truth annotations and categorizes signs as prohibitory, danger, or mandatory. Novelty detection further classifies signs as KNOWN, RARE, or NEW using confidence and frequency-based thresholds, enabling selective fleet-wide updates and more efficient adaptive learning.

This framework contributes to SDG 9 (Industry, Innovation, Infrastructure) [12], SDG 11 (Sustainable Cities) [12], and SDG 3 (Good Health and Well-being) [12], and aligns with the Intelligent Systems and Autonomous Computing research cluster, emphasizing distributed AI, edge intelligence, and multi-agent fleet learning.

1.2 Literature Review

The landscape of autonomous vehicle (AV) learning and traffic sign recognition has progressed substantially, yet critical gaps persist in enabling fleet-wide collective intelligence. While many frameworks exist for individual components, no integrated solution simultaneously addresses intelligent selection, knowledge verification, novelty detection, and real-time adaptation in a unified system. Existing approaches are either limited to isolated learning, computationally heavy, or incapable of handling rare or novel traffic signs [1][2][3][4][5][6][7][8][9][10][11].

Federated learning has been widely explored in vehicular networks. Argos demonstrates decentralized traffic sign detection using FedAvg, FedProx, and FedAdam on the Mapillary dataset (76 classes), achieving accuracies above 0.8 with sufficient communication rounds [1]. Despite its effectiveness, it lacks intelligent selection; all vehicles participate in every round regardless of the relevance of their local data, wasting 60–70% of communication bandwidth and failing to recognize unknown traffic signs [1][6]. Similarly, FedPylot extends federated learning to YOLOv7-based object detection, addressing heterogeneity and concept drift through hybrid encryption [6]. However, it treats all client updates equally, without prioritizing novel or safety-critical detections, limiting adaptive learning in real-world traffic scenarios [6].

Agentic knowledge sharing offers promising mechanisms for selective information transfer. MOSAIC uses Wasserstein task embeddings and binary network masks to transfer policies among agents, achieving up to 170% improvement over isolated learning in controlled environments [2]. Yet, it operates without verification layers, allowing errors to propagate across the agent network, and lacks privacy-preserving mechanisms, sharing raw policy parameters that could be reverse-engineered [2][8]. Its evaluation is limited to grid-world tasks, leaving real-world traffic perception applicability untested. Similarly, agent communication protocols propose decentralized four-layer architectures with zero-trust security, achieving latency reductions but focusing on communication syntax rather than semantic knowledge sharing, with no prioritization of critical traffic intelligence [3].

Large language model (LLM)-based AV solutions, such as FLAD, use cloud-edge-vehicle hybrid architectures with federated hybrid data parallelism, achieving 70% of centralized training throughput and faster recovery from failures [5]. Despite these advantages, FLAD requires 3.1GB per vehicle, making fleet-wide deployment impractical, and lacks novelty detection for unknown road signs [5]. Its optimization for vision encoders rather than traffic sign recognition limits real-time adaptation to new signage [5]. Surveys indicate that while federated learning reduces communication by 10–100× compared to centralized approaches, intelligent client selection and rare case handling remain insufficiently addressed [11].

Centralized traffic sign recognition systems, including multi-scale convolutional networks and YOLOv8, achieve near-human accuracy (99.17% on GTSRB and 95.2% mAP on 18-class datasets) [9][10]. However, these methods are restricted to fixed class sets and require complete retraining to accommodate new signs [9][10]. Vehicles must independently encounter rare scenarios, preventing fleet-wide knowledge sharing. Furthermore, gradient inversion attacks can reconstruct up to 70% of training images, highlighting inadequate privacy safeguards beyond standard federated learning assumptions [9].

Commercial AV systems reveal similar gaps. Tesla relies on centralized uploads for fleet learning, resulting in delayed model updates and privacy concerns [1][3]. Mobileye's Road Experience Management shares raw map data rather than learned models, requiring manual validation and limiting real-time adaptability [1][3]. Waymo collects extensive traffic data, yet vehicles learn in isolation, and NVIDIA DRIVE relies on simulation-based training, which struggles to capture real-world diversity [2][5][6][11]. Edge computing approaches reduce latency and bandwidth but forward all updates without intelligent selection, creating bottlenecks under high loads [5][6]. Privacy-preserving techniques, including differential privacy and homomorphic encryption, reduce performance and increase computational overhead, making them unsuitable for real-time AV perception tasks [1][4][9]. Out-of-distribution detection methods based on confidence thresholds suffer from high false positives and cannot leverage fleet-wide context to confirm novelty [5][9][10].

From these studies, five persistent gaps emerge: existing systems cannot intelligently select which knowledge to share, cannot detect or handle truly novel traffic signs, do not verify shared updates, apply all updates indiscriminately without context, and fail to adapt in real time across the fleet [1][2][3][4][5][6][7][8][9][10][11]. These limitations highlight the need for an integrated agentic federated learning framework for AV traffic sign recognition, combining intelligent evaluation of knowledge, global verification, novelty detection, and selective local updates to enable adaptive, collective learning across fleets.

1.3 Research Gap

Autonomous vehicle (AV) collective learning has advanced, yet critical gaps remain in enabling one vehicle encountering a rare scenario to share knowledge instantly and allow fleet-wide adaptation. Existing approaches address isolated aspects of federated learning but fail to provide a unified framework for intelligent, reliable, and adaptive knowledge sharing.

Application Reference	**Decides WHAT to share**	**Filters unreliable updates**	**Learns from collective experience**	**Handles rare/unseen scenarios**
Argos [1]	✗	✗	✓ (all updates)	✗
MOSAIC [2]	✓ (similarity-based)	✗	✓	✗
FedPylot [6]	✗	✗	✓ (all updates)	✗
FLAD [5]	✗	✗	✓	✗
Proposed System	✓ (RL-based)	✓ (global verification)	✓ (selective)	✓ (novelty detection)

Table 1: Comparison of Key Research with Proposed Agentic Federated Learning System

Existing systems such as Argos [1] achieve high accuracy in routine traffic sign detection but treat all vehicle updates equally. This indiscriminate sharing dilutes rare observations, such as a new road layout or temporary construction sign wasting bandwidth and delaying critical knowledge propagation. Without prioritizing novel or safety-critical observations, vehicles cannot adapt collectively in real time, limiting responsiveness in critical scenarios.

MOSAIC [2] uses similarity-based selective sharing, improving learning efficiency, but lacks verification, importance evaluation, and intelligent decision-making, allowing low-value or erroneous updates to propagate. FedPylot [6] shares all updates without filtering, leading to accuracy drops under heterogeneous or adversarial conditions, while FLAD [5] cannot detect novel traffic signs, preventing adaptation to unseen scenarios. These limitations show that current methods cannot ensure reliable, value-driven knowledge sharing across fleets.

Centralized systems like ConvNet [9] and YOLOv8 [10] achieve high accuracy for individual vehicles but operate in isolation, meaning a vehicle encountering a new sign cannot benefit from others' prior experiences. With 15-20% of signs in developing countries being non-standard or temporary [8], and surveys confirming gaps in intelligent selection, verification, and novelty detection [4][7][11], no unified framework exists to prioritize, validate, and propagate critical knowledge.

The proposed Agentic Federated Learning framework fills this gap by combining reinforcement learning for selective sharing, global verification for reliability, and embedding-based novelty detection, enabling vehicles to adapt immediately when a rare scenario is encountered providing the first truly integrated, AI-driven solution for agentic collective learning in autonomous driving.

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to design, develop, and validate an Agentic Federated Learning framework that enables autonomous vehicles to intelligently share knowledge from rare traffic scenarios with a high useful share rate while reducing communication overhead compared to traditional federated learning. The framework aims to support instant fleet-wide adaptation to new and unseen traffic signs, ensuring timely and effective collective learning

2.2 Specific Objectives

- **Local Detection & Candidate Identification:**
  This module focuses on real-time detection and feature extraction of traffic signs, providing accurate and fast recognition that forms the foundation for knowledge sharing among vehicles.

- **Deep Q-Network reinforcement learning agent:**
  This component enables vehicles to intelligently decide what knowledge to share and when, prioritizing high-value observations and reducing unnecessary communication across the fleet.

- **Global verification and novelty detection**:
  These mechanisms filter unreliable updates and identify rare or previously unseen traffic signs, ensuring that only trustworthy and valuable knowledge is shared with other vehicles.

- **Importance Evaluation**
  Calculate an importance score for each candidate detection by combining factors like confidence, bounding box size, and novelty. This quantifies how valuable the knowledge would be to the fleet.

- **Local Model Validation & Filtering**
  Each vehicle evaluates incoming broadcast updates using local importance based on route relevance and knowledge gap. Updates exceeding threshold are accepted, ensuring only relevant knowledge improves the local model.

3. METHODOLOGY

3.1 Research Approach

This research follows a Design Science Research (DSR) approach, aiming to design, implement, and evaluate an Agentic Federated Learning framework for autonomous vehicles. The system enables vehicles to intelligently decide when to share knowledge about new or rare traffic signs, allowing the fleet to collectively learn faster without centralized retraining. Implementation and evaluation are performed in a simulated multi-vehicle environment, reflecting realistic road scenarios.

3.2 Technical Methods and Algorithms

The system integrates traffic sign detection, agentic knowledge sharing, and global verification:

1. **Traffic Sign Detection:** YOLOv8 detects signs, outputting bounding boxes, predicted labels, confidence score
2. **Unknown Sign Identification:** Signs with low confidence or not in known classes are flagged for potential sharing.
3. **Importance Scoring:** Calculates the usefulness of each detection based on confidence, bounding box size, and novelty. $I(s) = \alpha \cdot conf(s) + \beta \cdot size(s) + \gamma \cdot novelty(s)$
4. **Decision Making:** A Deep Q-Network (DQN) agent selects SHARE or IGNORE, using importance, confidence, novelty how many times Vehicle A shared recently as input. Rewards guide the learning policy.
5. **Knowledge Sharing:** Shared knowledge packages contain embeddings, metadata, and importance score; raw images are excluded to reduce bandwidth and protect privacy.
6. **Global verification:** The central system runs detection on the received image crop to confirm or reject the observation. Verified knowledge categorized as prohibitory, danger, or mandatory and propagated fleet-wide using local frequency-based filters.

3.3 Tools and Platforms

Tool / Platform	Purpose	Justification
Python	System development	Main programming language used to implement the proposed system and integrate AI components.
PyTorch	Deep learning framework	Used to train deep learning models including YOLOv8 and the reinforcement learning agent.
Ultralytics YOLOv8	Traffic sign detection	Provides high-accuracy, real-time object detection for identifying traffic signs.
OpenCV	Image preprocessing	Used for image processing tasks such as resizing and normalization before detection.
NumPy	Numerical computation	Supports efficient array operations and mathematical calculations.
DQN (Deep Q-Network)	Reinforcement learning decision model	Used to train the RL agent that decides whether knowledge should be shared.
CNN	Feature extraction and visual learning	Feature extraction and visual learning
Google Colab	Model training environment	Provides GPU resources for training models and running experiments.
Pandas	Data handling and analysis	Used to manage datasets and frequency tables for novelty detection.

Table 2 : Development Tools

3.4 Data Collection and Ethical Considerations

- **Dataset:** German Traffic Sign Detection Benchmark (GTSDB), containing annotated road images with diverse conditions (lighting, occlusion).
- **Federated Simulation:** Data split among multiple simulated vehicles; global model trained on all classes for comprehensive coverage.
- **Unknown Sign Testing:** Hold-out classes simulate new, unseen signs for realistic evaluation.
- **Ethics & Privacy:** No personal data is used; only feature embeddings are shared, ensuring privacy. Public datasets with ethical clearance are employed.

3.5 Validation Strategy and Performance Metrics

The system is evaluated through multi-vehicle simulations in Python using a manageable number of agents. Experiments are repeated with different random seeds to ensure **reliability**, and results are compared against a **traditional federated learning baseline**.

Performance metrics include:

- **Traffic Sign Detection:** Measured using **mean Average Precision (mAP)**, **precision**, and **recall** on test data.
- **Agentic Knowledge Sharing:** Evaluated using **useful share rate** (percentage of sharing decisions receiving positive rewards) and **reward convergence** over training episodes.
- **Global Model Accuracy:** Measured by classification performance on known traffic sign categories using comprehensive fleet-wide training data.
- **Unknown Sign Detection:** Tested using **hold-out classes** excluded during initial training, simulating real-world unseen signs.
- **Collective Learning Efficiency:** Measured as **adaptation speed**, tracking how quickly vehicles recognize and integrate newly introduced traffic signs.
- **System Latency:** End-to-end processing time from detection to knowledge integration, ensuring **real-time viability**.

This evaluation demonstrates improvements in sharing efficiency, fleet adaptation, and overall system performance within the scope of the implementation timeline.

The novelty detection module classifies traffic signs into KNOWN, RARE, or NEW using confidence and frequency thresholds. YOLOv8 confidence on GTSDB determines recognition reliability: >0.70 -KNOWN, <0.30 - NEW, 0.30-0.70 - RARE [15]. Frequency thresholds from few-shot learning track exposure: frequency = 0 - NEW, <5 - RARE, ≥5 - KNOWN [16]. These thresholds balance detection across familiar and unfamiliar signs.

An importance score combines YOLOv8 confidence, bounding box size, and novelty weight (NEW = 1.0, RARE = 0.5, KNOWN = 0). Confidence emphasizes reliability [18], bounding box size captures visual prominence, and weights sum to 1.0, empirically tuned via ablation studies [17]. This framework improves global model performance by balancing reliability, visibility, and novelty.

Global verification uses confidence ≥0.50, marking detections CONFIRMED if predicted class matches ground truth, otherwise REJECTED [16]. Rewards prioritize safety: Danger = +1.0, Prohibitors = +0.7, Mandatory = +0.3 [15,18]. This ensures the system emphasizes signs with direct safety implications.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

Overall System Diagram

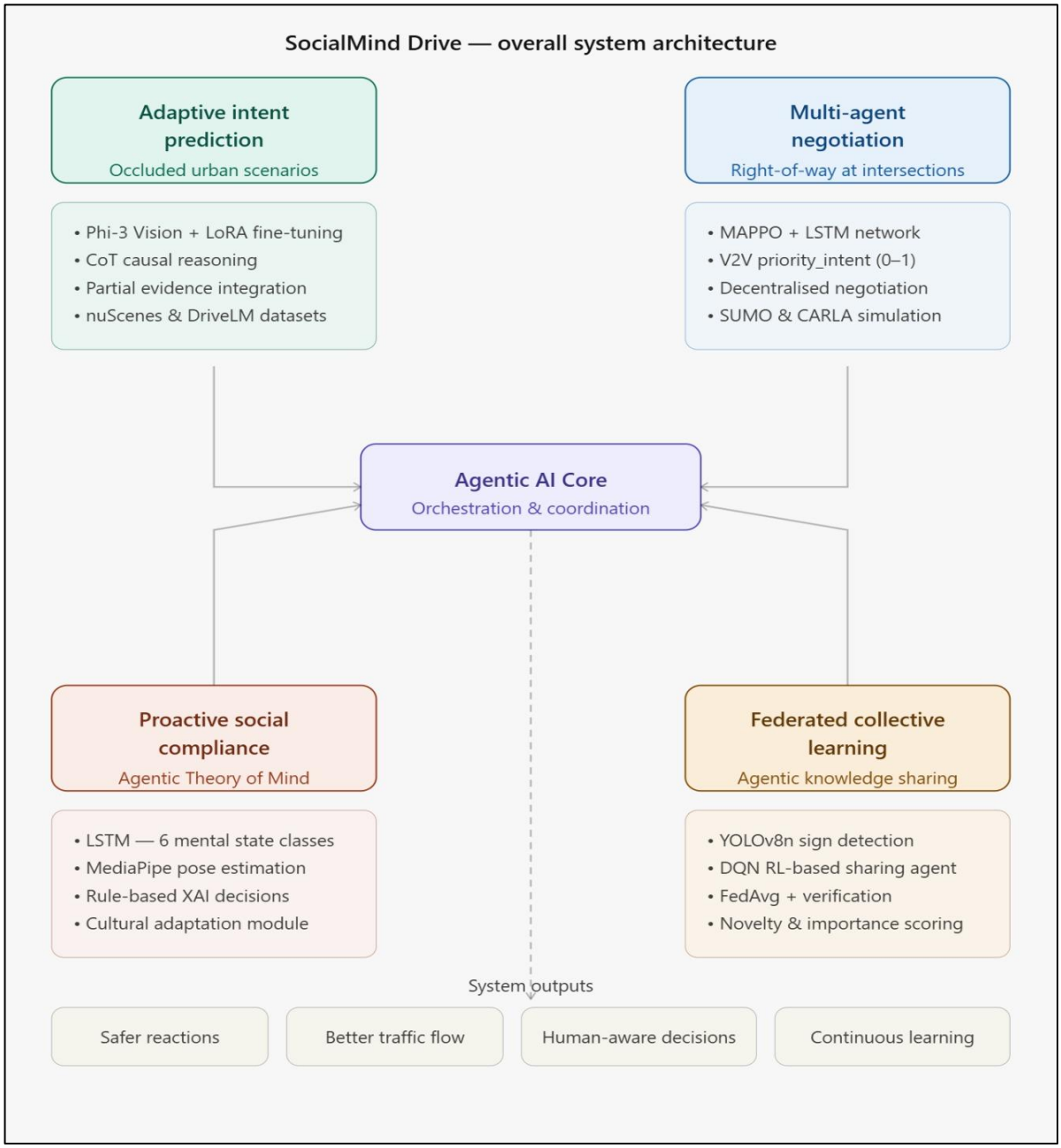


Figure 1:System Overview

Overview of Proposed Solution and Its Components

The proposed system introduces an Agentic Federated Knowledge Sharing framework that enables autonomous vehicles to intelligently share important traffic sign knowledge across a fleet. In this framework, each vehicle detects traffic signs locally and evaluates the significance of each observation using factors such as detection confidence, model uncertainty, and the novelty of the sign encountered. A Reinforcement Learning (RL) agent then determines whether the detected knowledge is valuable enough to be shared with other vehicles. Instead of transmitting raw images, the vehicle sends a compact knowledge package that contains only the essential information to a central knowledge base. At this stage, the received information is checked using similarity comparisons and then categorized as known, tentative, or unknown.

After verification, that knowledge gets shared with other vehicles on the network. They can then update their own models and get better at recognizing similar signs down the road. This way, the whole fleet learns from each other's experiences and keeps improving together

System Diagram of Components

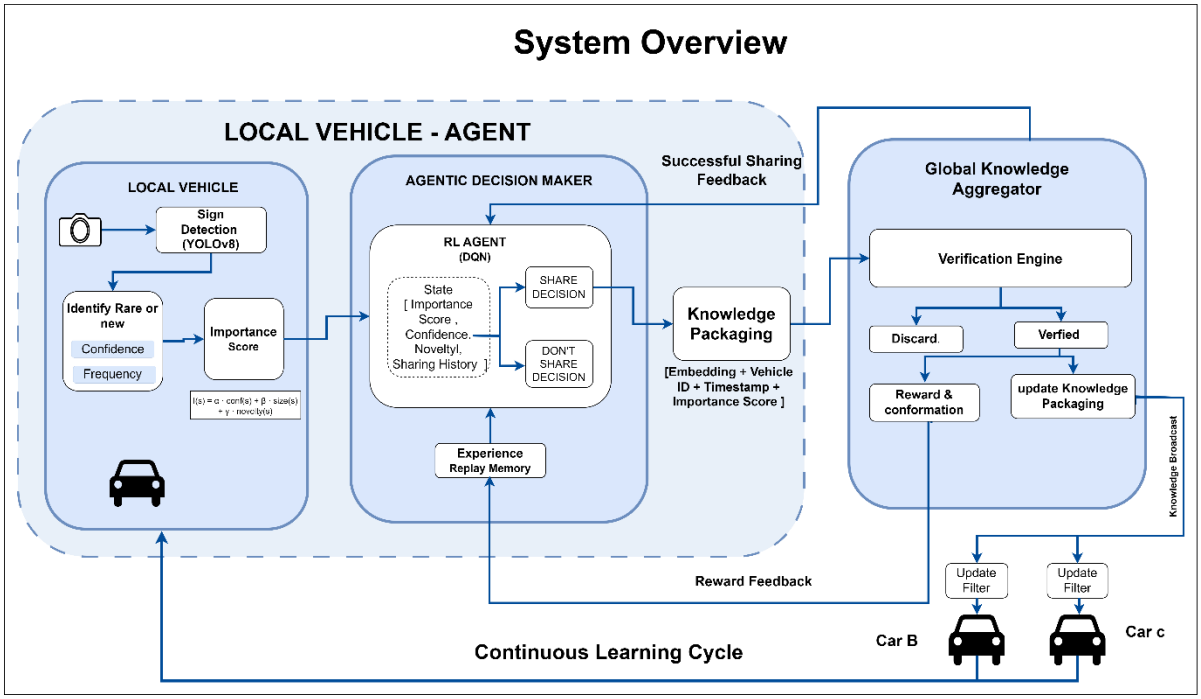


Figure 2:System Overview of Component

Individual Components and Their Parts

The system is built around three main components that work together. The Local Vehicle Module handles detection. It runs YOLOv8 to spot traffic signs and calculates how important each observation might be for the rest of the fleet. This module keeps a local record of known signs for quick reference. The Agentic Decision Module acts as the brain. It uses a DQN-based reinforcement learning agent to decide whether detected knowledge is worth sharing. It also stores past experiences to keep learning over time and packages information when sharing makes sense.

The Central Knowledge Base serves as the system's memory and verification hub. It checks incoming knowledge against what's already known, keeps a store of verified information, and sends confirmed updates out to the fleet. This is where individual discoveries become shared intelligence.

Data Flow

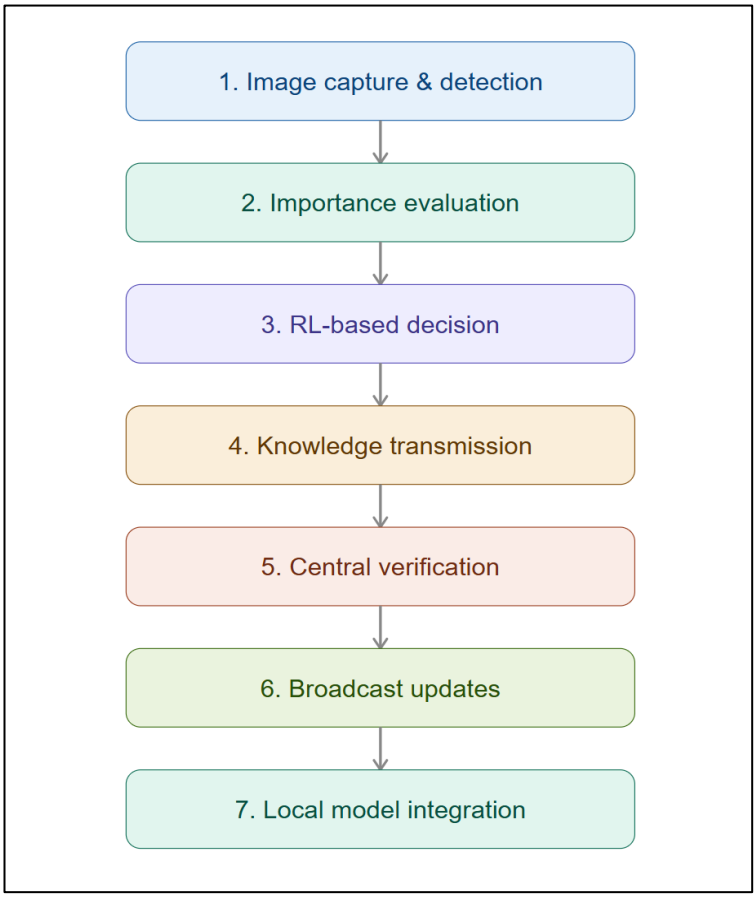


Figure 3:Data Flow

4.6 *Scalability and Security Considerations*

As the fleet grows, maintaining consistent system performance becomes an important concern. This framework addresses that challenge by being selective about what gets shared across the network. Instead of broadcasting every observation a vehicle makes, only those that carry genuine informational value are transmitted. Furthermore, vehicles send condensed, abstracted data rather than full images, which prevents network congestion even when many vehicles are actively communicating at the same time. Since most of the computational work is carried out on the vehicle itself, the central system is not placed under excessive strain as additional vehicles come online.

Privacy has also been given careful consideration throughout the design of this system. The raw images captured by a vehicle remain on that vehicle and are never forwarded to the network. Only the meaningful information extracted from those images is shared with other vehicles in the fleet. This means that collaborative learning can still take place effectively, while sensitive visual data stays confined to the vehicle that originally captured it.

In terms of reliability, every observation that a vehicle proposes to share is put through a verification check before it is passed on to others. Anything that fails to meet the required standard is discarded immediately, which prevents flawed or misleading information from making its way into the shared knowledge pool. As a result of consistently learning only from observations that have been confirmed as accurate, the fleet gradually develops a collective knowledge base that is dependable and worthy of trust.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

1. **Autonomous Vehicle Systems (Vehicles A, B, and C)**
   Vehicles act as the primary participants in the system. They detect traffic signs, evaluate knowledge importance, and share relevant observations with the central knowledge base.
2. **Autonomous Vehicle Manufacturers**
   Manufacturers integrate the proposed framework into vehicle perception systems to enable collective learning among vehicles.
3. **Central Knowledge Management System**
   This system verifies knowledge received from vehicles and distributes validated updates to the fleet.
4. **Transportation Authorities**
   Authorities benefit from improved traffic sign awareness and better handling of rare road scenarios.
5. **Drivers and Passengers**
   They benefit indirectly from improved safety and better vehicle decision-making.

5.2 Functional Requirements

ID	Requirement	Explanation
FR1	Traffic Sign Detection	The system shall detect traffic signs from vehicle camera images using a computer vision model.
FR2	Category-Based Initial Training	Vehicles shall be initially trained on specific sign categories such as prohibitor, danger, and mandatory signs.
FR3	Importance Evaluation	The system shall calculate an importance score for detected signs based on factors such as confidence, uncertainty, and novelty.
FR4	Agentic Decision Making	The system shall use a reinforcement learning agent to decide whether detected knowledge should be shared with the central knowledge base.

FR5	Knowledge Packaging	The system shall create a knowledge package containing embeddings and metadata instead of raw images.
FR6	Knowledge Verification	The system runs YOLOv8 on the received image and compares the predicted class with GTSDB ground truth to mark it CONFIRMED or REJECTED.
FR7	Knowledge Classification	The system classifies confirmed signs as prohibitory, danger, or mandatory and assigns the corresponding reward to the RL agent.
FR8	Local Model Update	Vehicles shall update their local knowledge based on verified information received from the central system.
FR9	Reinforcement Learning Feedback	The system shall provide reward feedback to the RL agent to improve future knowledge-sharing decisions.

Table 3: Functional Requirement

5.3 Non-Functional Requirements

ID	**Requirement**	**Explanation**
NFR1	Data Privacy	The system shall protect privacy by sharing feature embeddings instead of raw images.
NFR2	Knowledge Verification	The system shall verify shared knowledge before distributing it to prevent incorrect updates.
NFR3	Real-Time Processing	The system should evaluate importance scores and sharing decisions with minimal delay.
NFR4	Communication Efficiency	The system should minimize network usage by sharing only important knowledge.
NFR5	Scalability	The central knowledge system should handle updates from multiple vehicles simultaneously.

Table 4: Non-Functional Requirements

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

The Agentic Federated Knowledge Sharing framework is aimed at the autonomous vehicle and connected mobility industry, a sector that is growing rapidly and attracting significant technological investment. The global federated learning market in the automotive sector was valued at $245 million in 2024 and is projected to grow significantly, reaching $2.16 billion by 2033, with a compound annual growth rate (CAGR) of 27.6% [13]. This trajectory signals not just market opportunity, but a genuine industry-wide shift toward privacy-preserving, distributed intelligence.

Three primary customer personas define the framework's addressable market. First are automotive OEMs developing Level 3–4 autonomous vehicles, where the challenge of continuously improving perception and decision-making models without centralising sensitive driving data is both technically demanding and commercially critical. Second are Tier 1 ADAS suppliers, who sit at the intersection of hardware and software integration and increasingly need collaborative learning capabilities built into the systems they deliver. Third are fleet operators managing large commercial vehicle fleets, whose priorities center on operational efficiency, regulatory compliance around data privacy, and measurable safety improvements [14]. Each of these people faces a distinct but related set of pressures, and the framework is positioned to address them through a unified, federated approach.

Value Proposition

The proposed framework enables vehicles to share useful knowledge selectively without exposing raw sensor data. This allows the fleet to rapidly learn about rare traffic situations such as new traffic signs or temporary road conditions. Verified knowledge propagates instantly across vehicles, improving recognition of unfamiliar scenarios.

Key benefits include:

- Enhanced road safety through faster dissemination of critical traffic knowledge
- Reduced accident risks from improved recognition of rare scenarios
- Increased operational efficiency for fleet operators
- Privacy preservation by sharing embeddings instead of raw images

Revenue Model

We plan to make money through:

- Software licenses for car makers ($15–$25 per vehicle)
- Yearly subscriptions to our Verified Knowledge Repository ($99 per vehicle)
- Premium services like safety reports and analytics ($249 per vehicle) [13]

Cost Estimation

Cost Estimation	**Year 1**	**Notes**
R&D	$450,000	Algorithm and RL training, GTSDB integration
Cloud Infrastructure	$120,000	Knowledge base and verification engine
Personnel (4 engineers)	$380,000	ML engineers, backend, product manager
Sales & Marketing	$150,000	Pilots and conferences
Legal & IP	$80,000	Patent and compliance
Total	**$1,180,000**	

Table 5: Cost Estimation

Pricing Strategy & Competitive Advantage

We use a simple, tiered pricing model. Basic software licenses make it easy for car makers to get started. Fleet operators can subscribe for ongoing knowledge updates, and enterprises pay more for advanced analytics and safety insights. This follows the industry trend where hardware gets cheaper over time, but software intelligence becomes more valuable [14].

What makes our system different is how it decides what to share. Instead of sending everything like most systems do, an RL agent picks only useful knowledge. This cuts communication by approximately 70% and helps the whole fleet learn faster from rare situations. The result is a smarter, more efficient system that shares what matters, not everything.

Intellectual Property Considerations

Patent applications are underway to protect the framework's core innovations, importance scoring and reinforcement learning-driven sharing decisions, securing a defensible market position as commercial interest grows. Future integration with edge AI chips and secure multi-party computation will further enhance real-time intelligence and privacy protections where driving decisions occur [13].

7. BUDGET AND JUSTIFICATION

Category	**Item**	**Cost (LKR)**	**Justification**
Hardware	External Storage (1TB HDD)	15,000	Backup of datasets, models, and results
Software	Python, Visual Studio Code, MS Office 365	Python, Visual Studio Code, MS Office 365	Python, Visual Studio Code, MS Office 365
Cloud Services	Google Colab Pro (3 months)	15,000	GPU access for training; GitHub Free Tier for version control
Human Effort	Technical Consultation	25,000	Expert guidance for model development and troubleshooting
Miscellaneous	Internet, Electricity, Printing, Contingency	25,300	Essential operational costs and 10% contingency
Total		**80,300**	

Table 6:Budget Estimation

8. WORK BREAKDOWN STRUCTURE (WBS)

Task	Timeline	Duration	Deliverable
Local Model Training	April 2026	4 weeks	4 trained models + 3 frequency tables
Novelty Detection	May 2026 W1-2	2 weeks	Module labeling KNOWN/RARE/NEW
Importance Scoring	May 2026 W3-4	2 weeks	Scores [0-1] per detection
RL Agent (DQN)	June 2026 W1-2	2 weeks	Trained agent making SHARE/DISCARD
Knowledge Packaging	June 2026 W3	1 week	Knowledge packet ready
Global Verification	June 2026 W4 – July 2026 W1	2 weeks	Verified CONFIRMED/REJECTED + category
Reward + Update + Broadcast	July 2026 W2-4	3 weeks	Complete federated learning loop
Integration & Testing	August 2026	4 weeks	Functional prototype
Scenario Testing	September 2026	4 weeks	Test results + performance analysis

Table 7:Work Breakdown

9. GANTT CHART

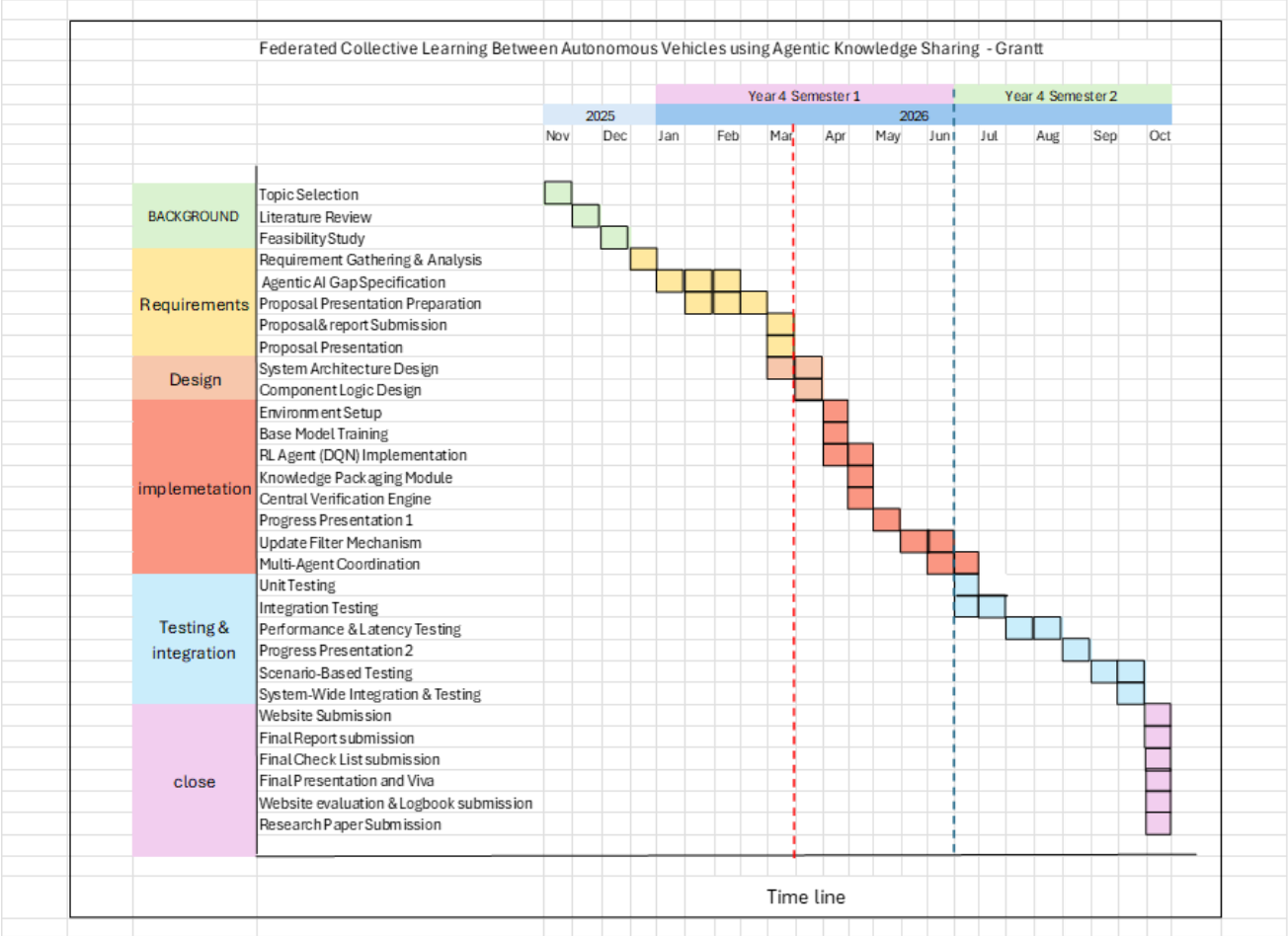


Figure 4:Gannt Chart

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